

Exam Logistics

- **Date / Time:** Week 6 (First 75 minutes of class).
- **Weight:** 25% of course total.
- **Scope:** Weeks 1, 2, 3, and 4.
- **Rules:** Closed-book, closed-notes. Simple non-programmable calculator allowed. No AI tools, laptops, or smartphones.

Exam Format

- **Section A:** Conceptual Multiple Choice / Short Questions (30%).
- **Section B:** Quantitative Problem Solving & Arbitrage Execution (50%).
- **Section C:** Policy & Risk Intuition Transfer (20%).

Core Concepts to Master

- Quotation convention: BASE/QUOTE (Base = 1 unit, Quote = price).
- Bid-Ask Rule: Dealer buys base at Bid, sells base at Ask. Customer trades on opposite side!
- Implied Cross-Rate calculation with bid-ask spreads.

[View Week 1 Sample Exam Problem & Solution →](#)

Triangular Arbitrage Steps

- ① Compare implied cross-rate with direct market quote.
- ② Trace 3-step currency loop (Clockwise vs. Counter-Clockwise).
- ③ Multiply execution rates (sell base at bid, buy base at ask). Profit if loop product > 1 .

Let S and F be home currency per foreign unit, with rates matched to horizon T .

- **CIP**: covered home and foreign investments have equal payoffs in the frictionless benchmark.

$$F^* = S \frac{1 + r_h T}{1 + r_f T}.$$

Executable arbitrage requires a discrepancy large enough to cover trading and financing costs.

- **UIP**: with zero expected excess currency return,

$$E_t[S_{t+T}] = S_t \frac{1 + r_h T}{1 + r_f T}.$$

Risk premia and expectation errors can produce observed excess returns.

- **PPP**: $S_{PPP} = P_h/P_f$. The real exchange rate is $q = SP_f/P_h$.

Carry Trade Strategy

- Borrow in low-yield funding currency (r_F low) and invest in high-yield target currency (r_H high), unhedged.
- If S is funding-currency units per target-currency unit, approximate expected excess return is $(r_H - r_F) + E[\Delta S/S]$.
- Positive historical average excess returns are sample-dependent and exposed to currency, crash, funding, and liquidity risk.

Crash Risk & Regimes

- **Tail Risk:** High-yield currencies crash sharply during market stress ("up the stairs, down the elevator").
- **Exchange Rate Regimes:** Fixed/Pegged vs Crawling vs Floating; Policy Trilemma trade-offs.

[View Week 3 Sample Exam Problem & Solution →](#)

Comparing the Four Hedging Strategies for a Foreign Receivable (CF^*):

Strategy	Payoff / Lock Formula	Key Operational Feature
1. Unhedged	$CF^* \cdot S_{t+1}$	Full upside participation, full downside risk
2. Forward Lock	$CF^* \cdot F_t$	Fixes the conversion rate for a certain amount/date; retains counterparty and settlement risk
3. Money Market	$CF^* \cdot \frac{1}{1+r_f} \cdot S_0 \cdot (1 + r_d)$	Idealized equivalent to a forward under CIP and matched funding rates
4. FX Option	$\max(CF^* K, CF^* S_{t+1})$ FV of premium	— Floor on downside, preserves appreciation upside

- ① **Quote Direction:** Always identify which currency is BASE and which is QUOTE before calculating.
- ② **Bid/Ask Side:** When selling base currency, use dealer's **BID**. When buying base currency, use dealer's **ASK**.
- ③ **Tenor Alignment:** Convert annual rates using the quoted instrument's day-count and compounding convention (e.g., Actual/360 where applicable).
- ④ **Cash Flow Timing:** Map out cash flow timing diagram before taking Present Value (PV).
- ⑤ **Economic Intuition:** Explain the economic mechanism behind your numerical answer in 1–2 clear sentences.

Example →

Problem: You observe the following quotes from two dealers: Dealer A: EUR/USD = 1.0820 – 1.0825. Dealer B: GBP/USD = 1.2640 – 1.2648. Dealer C: GBP/EUR = 1.1650 – 1.1658. Is there a profitable triangular arbitrage? **Solution:**

- ① Calculate implied GBP/EUR cross-rate from Dealers A and B:

$$\text{Implied Bid}_{\text{GBP/EUR}} = \frac{\text{Bid}_{\text{GBP/USD}}}{\text{Ask}_{\text{EUR/USD}}} = \frac{1.2640}{1.0825} = 1.16766$$

$$\text{Implied Ask}_{\text{GBP/EUR}} = \frac{\text{Ask}_{\text{GBP/USD}}}{\text{Bid}_{\text{EUR/USD}}} = \frac{1.2648}{1.0820} = 1.168946$$

- ② Compare Implied Cross-Rate (1.1677 / 1.1690) with Dealer C quote (1.1650 / 1.1658).
- ③ Dealer C's Ask (1.1658) is LOWER than Implied Bid (1.1677)! Buy GBP from Dealer C at 1.1658 EUR, sell GBP for USD at Dealer B (1.2640), buy EUR with USD at Dealer A's ask (1.0825).
- ④ Profit per EUR 1M traded = EUR 1,000,000 $\times$ $\left(\frac{1}{1.1658} \times 1.2640 \div 1.0825\right) - 1,000,000 = +\text{EUR } 1,602$ (before fees and size limits).

Problem: Spot $S_0 = 7.8000$ HKD per USD. 1-year Forward $F_1 = 7.8200$ HKD per USD. $r_{\text{HKD}} = 5.0\%$, $r_{\text{USD}} = 3.0\%$. Identify and execute the covered interest arbitrage using a USD 10M borrowing notional. **Solution:**

- ① Calculate Theoretical CIP Forward Rate F^* :

$$F^* = 7.8000 \times \frac{1 + 0.050}{1 + 0.030} = 7.8000 \times \frac{1.05}{1.03} = 7.9515 \text{ HKD per USD}$$

- ② Actual Market Forward $F_{\text{market}} = 7.8200$ is below $F^* \implies$ USD is cheap forward relative to the interest differential.

- ③ Arbitrage Execution Strategy:

- ① Borrow USD 10,000,000 at 3%; owe USD 10,300,000 in one year.
 - ② Convert to HKD at spot: $\text{USD } 10\text{M} \times 7.8000 = \text{HKD } 78,000,000$.
 - ③ Invest HKD at 5%; receive $\text{HKD } 78\text{M} \times 1.05 = \text{HKD } 81,900,000$.
 - ④ Buy USD 10.3M forward at 7.8200, costing $\text{HKD } 10.3\text{M} \times 7.8200 = \text{HKD } 80,546,000$, and use the USD to repay the loan.
- ④ Net Profit = $\text{HKD } 81,900,000 - \text{HKD } 80,546,000 = +\text{HKD } 1,354,000$ before spreads, credit, and balance-sheet costs.

Problem: Investor borrows JPY 100M at $r_{\text{JPY}} = 0.1\%$ per annum, converts to AUD at $S_0 = 100.00$ JPY per AUD, invests at $r_{\text{AUD}} = 4.1\%$. 1 year later, spot rate moves to $S_1 = 92.00$ JPY per AUD (AUD depreciates 8%). Calculate net return in JPY. **Solution:**

- ① Convert initial JPY 100M to AUD: $\frac{100,000,000}{100.00} = \text{AUD } 1,000,000$.
- ② Invest in Australia for 1 year: $\text{AUD } 1,000,000 \times 1.041 = \text{AUD } 1,041,000$.
- ③ Convert back to JPY at $S_1 = 92.00$: $\text{AUD } 1,041,000 \times 92.00 = \text{JPY } 95,772,000$.
- ④ Repay JPY loan: $\text{JPY } 100,000,000 \times 1.001 = \text{JPY } 100,100,000$.
- ⑤ Net Payoff = $\text{JPY } 95,772,000 - \text{JPY } 100,100,000 = -\text{JPY } 4,328,000$, or -4.33% of the borrowed amount. An equity return would require the investor's equity contribution.
- ⑥ **Takeaway:** The $+4.0\%$ interest differential was more than offset by AUD depreciation, illustrating unhedged carry downside.

Problem: A HK firm expects EUR 1M receivable in 180 days. $S_0 = 8.5000$ HKD per EUR, 180-day $F = 8.5842$ HKD per EUR. Annualized simple rates (Actual/360): $r_{\text{HKD}} = 4.0\%$, $r_{\text{EUR}} = 2.0\%$. Put Option Strike $K = 8.5000$ with premium HKD 0.10 per EUR. Compare net HKD received under Forward Hedge vs Money Market Hedge vs Option Hedge (if $S_{180} = 8.4000$). **Solution:**

① **Forward Hedge:** Receives EUR 1M $\times 8.5842 =$ HKD 8,584,200.

② **Money Market Hedge:**

- Borrow EUR today: $\frac{\text{EUR 1M}}{1 + 0.02 \times \frac{180}{360}} =$ EUR 990,099.
- Convert to HKD at $S_0 = 8.5000 \implies$ HKD 8,415,842.
- Invest at HKD rate: $\text{HKD } 8,415,842 \times (1 + 0.04 \times \frac{180}{360}) =$ HKD 8,584,158.

③ **Option Hedge** (if $S_{180} = 8.4000 < K = 8.5000$): Exercise option at $K = 8.5000$.

- Gross HKD = EUR 1M $\times 8.5000 =$ HKD 8,500,000.
- Future value of option premium = EUR 1M $\times 0.10 \times (1 + 0.04 \times \frac{180}{360}) =$ HKD 102,000.
- Net HKD received = HKD 8,500,000 $-$ HKD 102,000 = HKD 8,398,000.

Illustrative classroom example. All figures and assumptions are hypothetical.

A Hong Kong firm will receive USD 1M in 90 days. Dealer USD/HKD is 7.79 bid / 7.81 ask. The HKD interest rate is 4% simple, Actual/360.

- **Wrong side:** converting USD to HKD today would use 7.79, since the customer sells the base currency. The ask applies when buying USD.
- **Wrong horizon:** the 90-day interest factor is $1 + 0.04(90/360) = 1.01$.
- **Wrong timing:** today's bid applies to immediate delivery. Lock the future receivable with a forward or money-market hedge.
- **Units and intuition:** a lower HKD-per-USD rate means each dollar buys fewer HKD, reducing the unhedged receipt.

Interpretation

Check quote direction, trade side, interest horizon, and cash-flow date before calculating. State the economic meaning of the result.